



DEDAN KIMATHI UNIVERSITY  
OF TECHNOLOGY  
UNIVERSITY EXAMINATIONS 2025/2026  
THIRD YEAR SECOND SEMESTER EXAMINATION FOR BACHELOR OF SCIENCE  
DEGREE IN MECHATRONIC ENGINEERING  
EMT 3204 ELECTRICAL MACHINE DRIVES II

DATE: 10TH FEBRUARY 2026

8.30AM-10.30AM TIME: 2 HOURS

**INSTRUCTIONS**

1. This paper consists of FIVE questions.
2. You are required to answer THREE questions only.
3. Question **ONE** is compulsory.
4. Attempt ONE question from section A and another ONE question in section B.
5. Question ONE carries 30 marks and the others carry 20 marks each.

**Question One (30 marks)**

- a) With the aid of sketches, discuss the rotating magnetic field concept of a stator containing three coils, each  $120^\circ$  apart. [12 marks]
- b) With an aid of a diagram, explain the power flow in an induction motor. [5 marks]
- c) Explain the following terms
- |                             |          |
|-----------------------------|----------|
| i. Voltage regulation       | [1 mark] |
| ii. Speed regulation        | [1 mark] |
| iii. Slipping poles         | [1 mark] |
| iv. Power-factor correction | [1 mark] |
- d) Name and explain the two types of AC machines [3 marks]
- e) Name the four factors that affects the torque generated by real ac machines [4 marks]
- f) Explain the difference between a Single excited machine and Doubly excited machine [2 marks]

## SECTION A (Answer only ONE question)

### Question Two (20 marks)

- a) Illustrated with simplified phasor diagrams ignoring the effect of  $R_A$ , the behavior of synchronous generators operating alone under changing loads at:
- i. lagging power factor [3 marks]
  - ii. unity power factor [3 marks]
  - iii. leading power factor [3 marks]
- b) A 208-V, 45-hp, 0.8-PF-leading, Delta-connected, 60-Hz synchronous machine has a synchronous reactance of 2.5 ohms and a negligible armature resistance. Its friction and windage losses are 1.5 kW, and its core losses are 1.0 kW. Initially, the shaft is supplying a 15-hp load, and the motor's power factor is 0.80 leading.
- i. Sketch the phasor diagram of this motor, and find the values of  $I_A$ ,  $I_L$  and  $E_A$ . [4 marks]
  - ii. Assume that the shaft load is now increased to 30 hp. Sketch the behavior of the Phasor diagram in response to this change. [3 marks]
  - iii. Find  $I_A$ ,  $I_L$  and  $E_A$  after the load change. What is the new motor power factor? [4 marks]

### Question Three (20 marks)

- a) Using phasor diagrams, explain the behavior of synchronous motors under varying conditions of Field current. [6 marks]
- b) A 480-V, 50-Hz, Y-connected, six-pole synchronous generator has a per-phase synchronous reactance of 1.0 ohms. Its full-load armature current is 60 A at 0.8 PF lagging. This generator has friction and windage losses of 1.5 kW and core losses of 1.0 kW at 60 Hz at full load. Since the armature resistance is being ignored, assume that the  $I^2R$  losses are negligible. The field current has been adjusted so that the terminal voltage is 480 V at no load.
- i. Calculate the speed of rotation of this generator. [2 marks]
  - ii. Find the terminal voltage of this generator if the following are true –
    - It is loaded with the rated current at 0.8 PF lagging. [1.5 marks]
    - It is loaded with the rated current at 1.0 PF. [1.5 marks]
    - It is loaded with the rated current at 0.8 PF leading. [1.5 marks]
  - iii. Find the efficiency of this generator (ignoring the unknown electrical losses), when it is operating at the rated current and 0.8 PF lagging. [3 marks]
  - iv. Find the voltage regulation of this generator at :
    - 0.8 PF lagging [1.5 marks]

- Unity PF .
- 0.8 PF leading .

[1.5 marks]

[1.5 marks]

### SECTION B (Answer only ONE question)

#### Question Four (20 marks)

- a) Discuss three basic approaches that can be used to safely start a synchronous motor.

[9

marks]

- b) A 480-V, 60-Hz, 50-hp, three-phase induction motor is drawing 60 A at 0.85 PF lagging. The stator copper losses are 2 kW, and the rotor copper losses are 700 W. The friction and windage losses are 600 W, the core losses are 1800 W, and the stray losses are negligible. Find the following quantities:

- The air-gap power,  $P_{AG}$  [2 marks]
- The power converted,  $P_{conv}$  [2 marks]
- The output power,  $P_{out}$  [2 marks]
- The efficiency of the motor [2 marks]

- c) State and explain the two different types of induction motor rotors

[3 marks]

#### Question Five (20 marks)

- a) The following test data were taken on a 7.5-hp, four-pole, 208-V, 60-Hz, design A, Y-connected induction motor having a rated current of 28 A.

|                           |  |                          |  |
|---------------------------|--|--------------------------|--|
| DC test:                  |  | $I_A = 8.20 \text{ A}$   |  |
| $V_{OC} = 13.6 \text{ V}$ |  | $I_C = 8.18 \text{ A}$   |  |
| No-load test:             |  | Locked-rotor test:       |  |
| $V_L = 208 \text{ V}$     |  | $V_L = 25 \text{ V}$     |  |
| $I_A = 8.12 \text{ A}$    |  | $I_L = 28.1 \text{ A}$   |  |
| $f = 60 \text{ Hz}$       |  | $f = 15 \text{ Hz}$      |  |
| $P_{in} = 420 \text{ W}$  |  | $P_{in} = 920 \text{ W}$ |  |
|                           |  | $I_A = 28.0 \text{ A}$   |  |
|                           |  | $I_C = 27.6 \text{ A}$   |  |

Sketch the per-phase equivalent circuit for this motor.

[16 marks]

- b) State and explain the four conditions to be met before paralleling synchronous generators.

marks]

[4